Week 03 - Tutorial 01

Transcript Language

English

Hello guys, today we will discuss about the subspace, what is subspace.

Suppose V is a vector space, then subspace is actually subset of vector space V with

W having some property.

So, let us go through by formal definition.

Suppose W is a subset of a vector space V over the, V is vector space over R, we call

W is a subspace of V, if W is also a vector space over R with and this vector space form over some operation and that operation will be same whatever the operation on V is there.

Now, we know that what is subset.

So, suppose it is V is a set then some collection of element of V, if you take that then if

you form a set of those elements then we will say that set is actually a subset of V and

what is vector space?

So, we know that suppose V is a set and we define two operation addition and scalar multiplication

on V with this addition and scalar multiplication having closed properties, suppose x and y

belongs to V then if you add them, add them and if it should V belongs to V and this scalar

multiplication.

So, suppose C is a real number and x is an element of V and this x, C times V also should

belongs to V.

If suppose these properties followed in V and with as we have seen in video lecture,

there are eight properties which we use to say oximes, if those properties followed by this operation, then we will say V is a vector space.

Now, suppose W is a subset of V and if we define the same operation addition and scalar

multiplication with the property same like this, the x belongs to W and Y belongs to

W and to making a subspace to, subspace of V it should be x plus y also belongs to W

and suppose C is a real number and x belongs to W then Cx should also belongs to W. This

is closed under addition and this is closed and this is called closed under scalar multiplication.

With same properties, this eight properties which V follows, if W will also follow then

we will say W is a subspace of V.

Now, suppose W is a given subset of V, then how we will check that W is also a subspace.

W is a subset of, it will be given then how we will check that W is subspace of V then

there is three criteria.

If W will satisfy that then we will say W is subspace of V. Those three criterias are

0 should belong to W and the second criteria is if x suppose x an element belongs to W,

y is an element of W then addition should also so belongs to W.

This is also called closed under addition and the third criteria is, suppose c is a

real number, and x is an element of W then their multiplication which is called a scalar

multiplication and if that one, that multiplication is also element of W, which is called a closed

under scalar multiplication then these, if these criteria followed by W, then we will

say W is a subspace of V.

Now, let us do some example.

Let the given vector space is R2 over R and we have given a subset W, W is x, y coordinate

and this, if we add x plus y it will become 0 and x, y element of R, these two conditions.

So, W is a subset given of R2.

We need to check W is a subspace or not.

So, let us our first criteria that 0 should belongs to W.

So, if we take x is equal to 0 and y is equal to 0, and if we add them 0 plus 0 it becomes

0, it means, 0, 0 belongs to W that means 0 element of R2 which is also belongs to W.

Now, we need to check the second criteria, the second criteria is let W1 is one element

is x1, y1 and the second W2 second element in x2 y2.

Now, if we add them w1 plus w2, so it will become x1, y1 plus x2, y2 this is equal to

coordinate y addition will happen same operation, as we did in R2.

So, this will become the x1 plus x2, y1 plus y2.

Now, w1 is element of W, W2 is element or W, it means, this two will satisfy this condition

means x1 plus y1 is equal to 0, and this is the for W1 and for W2, X2 plus Y2 is equal

to 0.

So, these two conditions we have given because W1 and W2 two are element of W. So, now, what

we did in addition of W1 plus W2 we got these coordinates x1 plus x2, y1 plus y2.

So, let us check is this addition is belongs to W, so we need to add them plus y1 plus y2 we need, we added them.

And we, as we said these are the element of R2.

So, that swapping can we do easily so, we can write x1 plus y1 plus x2 plus y2 and this

is 0 and this is 0 as w1.

This is 0 as w1 is that coordinate addition is becomes 0 as given condition and this is

0, so 0 plus 0 becomes 0.

It means, this addition belongs to W. So, second criteria is followed.

Now, we need to check third criteria.

The third criteria is a real number C and suppose W1 is equal to x1, y1 an element of

W and so W1 is element W it means x1 plus y1 is equal to 0.

Now, let us do multiplication C into W1 is equal to it will become coordinate wise multiplication

will happen, same operation as in R2.

So, it will become Cx1, Cy1.

Now we need to check this coordinates make sum 0 or not, then we will say this is scalar

multiplication is in W happened that is closed under scalar multiplication.

So, let us do the sum.

So, sum x1 plus Cy1 this we can write x1 plus y1.

We are able to do this because as we can able to do in R2, so we can do this and we have

given and this is 0 it means this is 0.

It means C times W1 is element of W. So, all three criterias done by W it means W is a

subspace of R2.

So, let us see it geometrically what do we mean, what we mean by a subspace.

So, this is the vector space R2 representing this plane is actually x, y, plane R2 means

x, y plane.

Now, we have given a subset W, which is coordinates x, y and having some condition x plus y is

equal to 0.

If you see it as in x, y plane so actually x this W represent a line, which pass through

origin with having slope minus 1.

So, actually W is this line which pass through origin and we if we see clearly the first condition 0 it passes through origin, so 0 element is here is present in W and if we

do addition with any taking to, any two elements then it will be the same in the line and again

and the third criteria is by taking any real number.

If you multiply with this any element of W, then it will be again in the line.

So, it means W is a subspace of R2.

Now, suppose, another W1 is a subspace of R2 and which having condition the same coordinates

with changing the slope actually.

So, subspace of R2 will be the passing represent the line, which passes through origin.

Why this is passed through origin because of the first criteria, 0 element would be

in W. So, any subspace of R2 other than trivial subspaces, which are 0 subspace and R2 itself,

represent a line which pass through origin.

Now, we will see a subset of R2, which will not form a subspace, it means, we have to

check those three criteria, which we discussed earlier and we will see that, which conditions will not followed by this W1.

So, let us check this, the first criteria that 0 element in W1.

So, first see clearly W1 is actually this subset of R2 with the same condition, as we

discussed earlier x plus y is equal to 0, but that x is that x and y belong, taken from

this integer that means x, y are integer.

So, we know that 0 belongs to z, so we can take the coordinate 0, 0 and this follow the

condition 0 plus 0 which is 0.

So, it means 0 element is in W. So, first criteria satisfied W1, now we need to check

second criteria.

The second criteria is, you will take two element from W1.

Let us take x1, y1 and second element is the x2, y2.

So, if we from W1 if we add them this will become coordinate wide addition and addition

of two integer is integer.

Also previously we have seen that addition of coordinates that is the x1 plus x2 plus

y1 plus y2 is equal to 0, here, same thing will also happen.

So, conditions of W1 are satisfied, it means it follow the addition, this W1 is closed

under addition.

So, it means this belongs to W1.

So second criteria followed by W1.

Now, the third criteria is, any real number multiplied with an element, which is at a

scalar multiplication and this is lacking W1.

So, let us take any real number we can take.

So, let us take any 1 by 2, this is a rational number and an element of W1 which is we can

take 1, n minus 1.

So, this is from W1 satisfying this condition and also these are the integer.

So, this is the element of W1.

Now, if we multiply them, so this is, this will be a coordinate wise multiplication,

so this will become 1 by 2 minus 1 by 2.

Now, here is the missing part that this is not an integer, this is not an integer, it

means W1 and is not closed under scalar multiplication.

So, here the third condition is lacking for W1.

So, the W1 is a subset, but not a subspace.